

Appendix M: The “M” Problems

MA Problems: The Basics

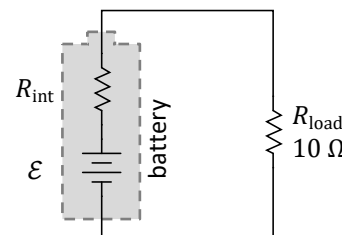
Answers to MA problems are on page A-18. **Updated Tuesday January 27, 2026 8:29pm.**

MA1.* If you haven’t done so already, please read the following documents on Blackboard under “Course Information:”

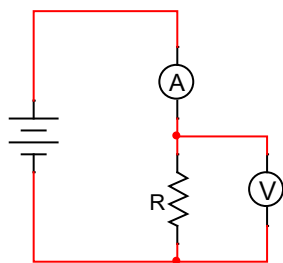
- The Syllabus
- “On Working Together Honestly”
- About Lab Notebooks
- Why no equation sheets

(a) If you could change any aspect of these course policies (or anything else about the course, I guess), what would you change? You must make at least one suggestion for a change; no fair just answering that you’re fine with whatever’s currently there. (b) How serious are you about this request? (Are you just suggesting a change because it was required, or because you genuinely think it would improve the course?)

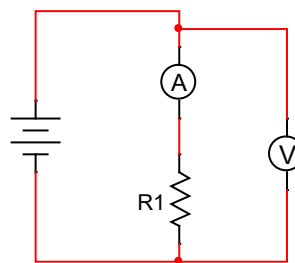
MA2. A student wants to find the internal resistance of a standard, size-AA battery. With no load resistor connected, the voltage across the terminals of the battery is 1.55 V. But when the battery is connected to a load resistor of $10\ \Omega$ (as shown on the right) the voltage across the load resistor is 1.47 V. What is the internal resistance of the battery?



MA3. The figures below show two possible circuits you could use to measure the current and voltage of a resistor. Both methods are imperfect, due to the non-ideal input impedances of the voltmeter and the ammeter. (a) Which circuit will give a current reading that is slightly inaccurate, due to the non-ideal voltmeter? (b) Will that current reading be too high or too low? (c) Which circuit will give a voltage reading that is slightly inaccurate, due to the non-ideal ammeter? (d) Will that voltage reading be too high or too low? (e) Suppose the impedances of the two meters are $10\ \Omega$ and $10\ \text{M}\Omega$. Which circuit is preferable if the resistor you are measuring has $R = 100\ \Omega$, and why? (f) Which circuit is preferable if the resistor you are measuring has $R = 1\ \text{M}\Omega$, and why?

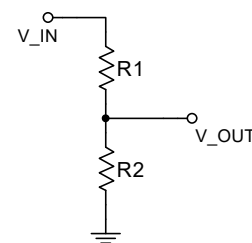


Circuit #1



Circuit #2

MA4. As shown in the figure to the right, two resistors in series may be thought of as a *voltage divider*, where the output voltage V_{out} is a fixed fraction of the input voltage V_{in} . Derive a general expression for V_{out} in terms of R_1 , R_2 , and V_{in} . Note that the word *derive* means to demonstrate using some mixture of mathematical statements along with some words and possibly a picture.



MA5. Consider a power supply like the battery depicted in problem MA2, in which the supply has an internal voltage ε and an internal resistance R_{int} , and where the supply will be connected to a load resistance R_{load} . (Feel free to call the resistances “ R_1 ” and “ R_2 ” if you prefer.) (a) Write an expression for the power P_{load} dissipated in the load resistor, in terms of only ε , R_{int} , and R_{load} . (b) For a supply with a fixed, unchangeable R_{int} , what value should you choose for R_{load} to *maximize* the power dissipated in the load? (c) For a fixed, unchangeable R_{load} what value would you want to choose for R_{int} to *maximize* the power dissipated in the load? (d) Based on your answer to part (c), it turns out that some battery types are just better at delivering high load power than others. Do a little Googling to find how the internal resistance of lithium ion batteries compares to, say, alkaline batteries. Are any other battery types that stand out as particularly good or particularly bad?

MA6. Students often wonder why resistors have the odd values that they often do: for instance, why 2.2 k Ω and 4.7 k Ω instead of 2 k Ω and 5 k Ω ? To find out why, check out the Wikipedia entry entitled “E series of preferred numbers.”

- Which set of numbers do the resistors in your kit follow? (E3? E6? E192?)
- In a single sentence, how are the numbers in that specific sequence calculated?

With the resistors that you have, it is almost always possible to combine just two of them either in series or in parallel to generate virtually any equivalent resistance you want to an accuracy of better than 1%. There are various handy online calculators to help you do so, such as the one at <https://svajkaj.com/>. Use this online calculator (or a different one, I guess) to calculate the following:

- What is the best way to approximate 7.0 k Ω using just two E6 resistors?
- What is the best way to approximate 2.7 k Ω using just two E6 resistors?

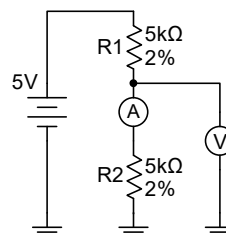
MA7. A student has connected the circuit below to their 5 V power supply, expecting a voltmeter reading of $V = 2.5$ volts. Which of the causes listed on the right could lead to each of the symptoms on the left? More than one answer may apply.

Symptoms

- The voltmeter reads 2.58 volts.
- The voltmeter reads 2.76 volts.
- The voltmeter reads 4.98 volts.
- The voltmeter reads 1.62 volts.
- The voltmeter reads 0 volts.

Causes

- The two leads of R_1 are touching each other (“shorted”).
- The two leads of R_2 are touching each other (“shorted”).
- The current meter is on the μA scale.
- R_2 has not been connected to ground.
- The current meter has a blown fuse.
- The voltmeter has a blown fuse.
- One side of R_2 has been connected to the 5 V supply.
- R_1 is actually a 10 k Ω resistor.
- The circuit is built correctly and the output is within the range expected for the tolerances of the components.



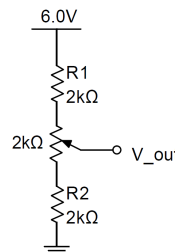
MA8. A student has constructed the circuit to the right, expecting an adjustable output voltage V_{out} in the range of 2 V to 4 V. Which of the causes on the right *could* lead to each of the symptoms on the left? More than one answer may apply.

Symptoms

- V_{out} range is 2 V to 4 V.
- V_{out} is a constant 6 V.
- V_{out} is a constant 4 V.
- V_{out} range is 3 V to 6 V.
- V_{out} range is 2 V to 3 V.
- V_{out} range is 1 V to 2 V.

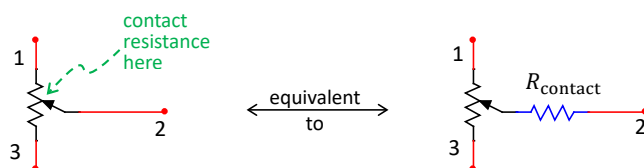
Causes

- R_2 has not been connected to ground.
- The two leads of R_1 are touching each other.
- The two leads of R_2 are touching each other.
- V_{out} connected to pin 1, not pin 2.
- Pins 2 and 3 are connected (“shorted”) together.
- V_{in} connected to 3 volts.
- The circuit is correct and the output is within the range expected for the tolerances of the components.



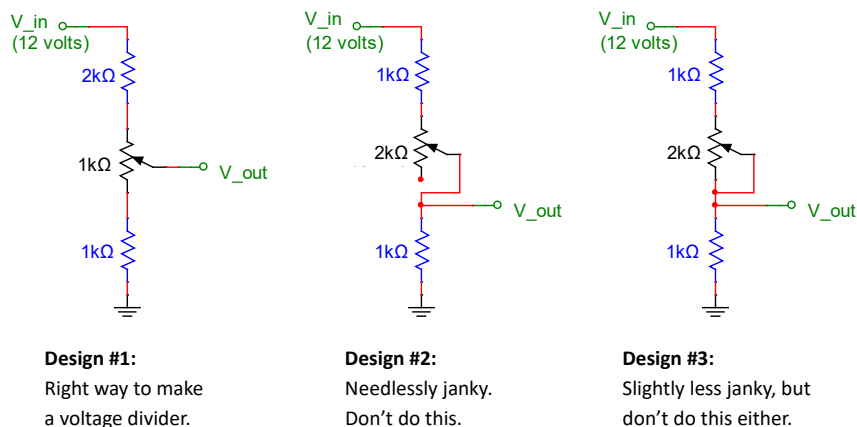
MA9. (a) Design a voltage divider that will guarantee that $V_{\text{out}}/V_{\text{in}} = 0.500$ using a trimpot and two fixed resistors: a $15\text{ k}\Omega$ and $15\text{ k}\Omega$ both 2% tolerance. (b) Design a voltage divider that will guarantee that $V_{\text{out}}/V_{\text{in}} = 1/3$ using a trimpot and two fixed resistors: a $100\text{ k}\Omega$ and $200\text{ k}\Omega$ both 1% tolerance. For both parts, use the minimum value of trimpot that will guarantee the correct output. (Note that “Design” means to draw the circuit diagram showing the values of all components.)

MA10. A common problem with trimpots and other potentiometers is that the moveable wiper (pin 2 in the drawing below) doesn’t always make good electrical contact with the resistive material between pins 1 and 3. The resulting *contact resistance* is particularly pernicious because it can fluctuate wildly over time or when the trimpot is adjusted, leading to effects that are both undesirable and unpredictable.



In the drawing below, the standard way to make a voltage divider is shown in Design 1, on the left.

- What is the ideal output voltage range for Design 1, assuming that the output is connected to an infinitely large impedance ($R_{\text{load}} = \infty$)? (in practice, the output of a voltage divider is almost always connected to something with a high input impedance—think of your voltmeter with $R_{\text{load}} = 10\text{ M}\Omega$, for instance.)
- What is the actual output range for Design 1 if the output is connected to a load with $R_{\text{load}} = 1\text{ M}\Omega$?
- How does the actual output range in (b) change with an additional contact resistance $R_{\text{contact}} = 1\text{ k}\Omega$?



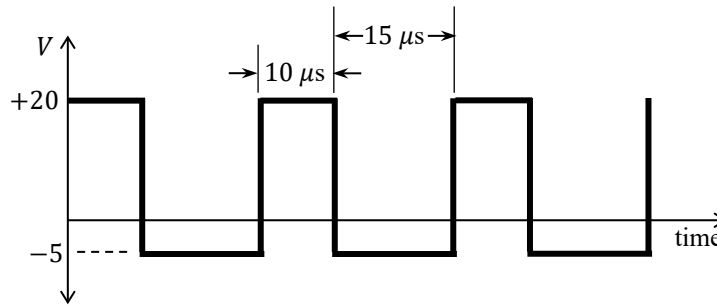
Your answers for parts (b) and (c) demonstrate that Design 1 is not affected much by contact resistance. Designs 2 and 3 above are also ways to build a voltage divider, but you’ll see now that these are much more sensitive to changes in contact resistance.

- Convince yourself that with $R_{\text{contact}} = 0$, Designs 2 and 3 give the same output range as Design 1. You can assume here and for the remaining parts of this problem that $R_{\text{load}} = \infty$.
- What is the output range for Design 2 with $R_{\text{contact}} = 1\text{ k}\Omega$? What about with $R_{\text{contact}} = \infty$?
- What is the output range for Design 3 with $R_{\text{contact}} = 1\text{ k}\Omega$? What about with $R_{\text{contact}} = \infty$?
- Why is Design 3 better than Design 2?

In general, you want as little current as possible to flow through the wiper of a potentiometer, to minimize the effect of contact resistance. You very rarely need to use a potentiometer as a two-terminal variable resistor, but if you do you should always connect the wiper lead to one of the outer leads as in Design 3.

MA11. A sinusoidal voltage with an amplitude of 10 V and a frequency of 0.25 Hz is placed across a load resistance of $220\ \Omega$. (a) What is the maximum *instantaneous* power dissipated in the resistor? (b) What is the *average* power dissipated in the resistor?

MA12. The periodic waveform shown below is 20 V for $10\ \mu\text{s}$ and $-5\ \text{V}$ for $15\ \mu\text{s}$.



- What range of voltages is displayed on an oscilloscope if the signal is measured using DC coupling?
- What range of voltages is displayed on an oscilloscope if the signal is measured using AC coupling?
- What is the RMS value of this signal as measured using DC coupling?
- What is the RMS value of this signal as measured using AC coupling?

MA13. (a) Find the RMS voltage of a periodic sawtooth wave which rises linearly from 0 to V_{max} and falls instantaneously to zero, such that $V = V_{\text{max}}t/\tau$, where τ is the period. Yes, you will need to compute an integral to answer this question. (b) Use your result from part (a) to find the RMS voltage of a triangle wave (as from your function generator) that oscillates between $-V_{\text{peak}}$ and $+V_{\text{peak}}$.

Answers to Selected MA Problems:

MA2. $0.544\ \Omega$ The circuit is built correctly and the output is within the range expected for the tolerances of the components. The circuit is built correctly and the output is within the range expected for the tolerances of the components. The circuit is built correctly and the output is within the range expected for the tolerances of the components.

MA3. (a) circuit #1 (c) circuit #2 (d) too high (e) circuit #1

MA4. $V_{\text{out}} = V_{\text{in}} \left(\frac{R_2}{R_1 + R_2} \right)$. (Hint: start by finding the current through the resistors.)

MA5. (b) $R_{\text{load}} = R_{\text{int}}$ (c) The correct answer is NOT $R_{\text{int}} = R_{\text{load}}$.

MA7. Of the nine lettered causes (a) thru (i), all but one gets used, and none are used more than once. Feel free to check your answers for each symptom 1 thru 5 with me; I will tell you what's wrong and let you try again.

MA8. Each symptom 1 through 6 has exactly one cause from the list. Feel free to check your answers for each symptom with me; I will tell you what's wrong and let you try again.

MA9. (a) $600\ \Omega$ (b) $2\ \text{k}\Omega$

MA10. (a) 3.0 to 6.0 volts. (b) 2.997751686 to 5.994005994 volts. (R_{load} changes the output voltages by about 0.1%.) (c) 2.997753931 to 5.994011976 volts. (R_{contact} changes the output voltages by about 0.0001%.) (e) 2.4 to 4.0 volts for $R_{\text{contact}} = 1\ \text{k}\Omega$, or constant 0 V for $R_{\text{contact}} = \infty$. (f) 3.0 to 4.62 volts for $R_{\text{contact}} = 1\ \text{k}\Omega$, or constant 3 V for $R_{\text{contact}} = \infty$. (g) With contact resistance, output of Design 3 is always closer to intended range than Design 2. In fact, the output of Design 3 will never be *outside* of the intended range, though the full intended range may not be accessible.

MA11. (a) 0.45 W (b) 0.23 W

MA12. (b) -10 to $15\ \text{V}$ (c) $13.23\ \text{V}$ (d) $12.25\ \text{V}$

MA13. (a) $0.58V_{\text{max}}$